



GEORG-AUGUST-UNIVERSITÄT
GÖTTINGEN IN PUBLICA COMMODA
SEIT 1737

Fakultät für
Physik



Master's Thesis

Improving the Reconstruction of Top-Quark Pairs in Dileptonic Final States in ATLAS Using Machine Learning

prepared by

Siemen Henning Aulich

from Emden

at the DESY Hamburg

ReferenceNumber: II.Physik-UniGö-MSc

Thesis period: 7th April 2025 until 30th June 2026

First referee: Prof. Dr. Stan Lai

Second referee: Dr. Katharina Behr

Abstract

Many of the recent highlights in particle physics research are related to top-quark physics. These include both the stringent tests of spin correlations and quantum effects in pairs of top quarks ($t\bar{t}$), and the observation of a possible quasi-bound state resonance in the $t\bar{t}$ invariant mass spectrum. The latter presents a very exciting opportunity to extend our current understanding of non-relativistic Quantum Chromodynamics. Both effects are predominantly studied in the dilepton decay channel of the top quark in a mass range close to the $t\bar{t}$ production threshold.

The dilepton final state contains two neutrinos, besides two charged leptons and two b -quark jets. Probing this system requires a precise reconstruction of the top quarks, which is complicated by the presence of the two neutrinos. While analytical regression strategies primarily focus on inferring the neutrino momenta, the problem of correctly assigning b -jets to their parent top quarks remains largely understudied. However, many of the sensitive variables used in $t\bar{t}$ precision measurements depend critically on the correct assignment of the jets.

Inspired by the success of machine learning architectures in tackling the assignment challenge for hadronic decay channels, this work investigates using a transformer model for the dilepton channel. An architecture specifically tailored to this channel's topology is shown to outperform all existing methods in assigning reconstructed particles to their parent partons, improving the resolution of relevant physics observables. Furthermore, it is investigated how these efforts can be combined with neutrino regression methods to offer a full reconstruction pipeline. Applying these methods to existing and upcoming analyses promises further enhancements in the sensitivity and precision of searches and measurements alike.

Keywords: Top Quark, Dileptonic Decay, Event Reconstruction, Machine Learning, Transformer Networks

Contents

1. Introduction	1
2. Theoretical Background	3
2.1. The Standard Model of Particle Physics	3
2.1.1. Electromagnetic Interaction	4
2.1.2. Weak Interaction	4
2.1.3. Strong Interaction	5
2.2. Top-Quark Physics	5
2.2.1. Spin Correlations in Top-Quark Pairs	7
2.2.2. Top-Quark Pair Quasi-Bound-State Effects	8
2.2.3. Dileptonic Decay Channel	8
2.2.4. Spin-Sensitive Angular Variables	9
3. Machine Learning in Particle Physics	11
3.1. Supervised Learning	11
3.1.1. Monte Carlo Training Data	11
3.1.2. Training	12
3.1.3. Validation and Testing	13
3.2. Regression and Conditional Density Learning	13
3.2.1. Supervised Learning as Risk Minimisation	13
3.2.2. Regression	13
3.2.3. Conditional Density Learning	14
3.3. Neural Networks	15
3.3.1. Dense Neural Network	15
3.3.2. Recurrent Neural Networks	16
3.3.3. Transformer Networks	17
3.3.4. Training Neural Networks	18
4. Experimental Setup	20
4.1. The LHC	20
4.2. The Detector	20
4.2.1. The Coordinate System	20
4.2.2. Inner Detector	21
4.2.3. Calorimeter System	22
4.2.4. Muon Spectrometer	23
4.2.5. Trigger System	23

Contents

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events	24
5.1. Data and Simulation	24
5.1.1. Simulation of Dileptonic Top-Quark Pair Events	24
5.1.2. Object Definition	24
5.1.3. Event Selection	25
5.2. Reconstruction of Top-Quark Pairs	25
5.2.1. B-Tagging	25
5.2.2. Jet Assignment Methods	26
5.2.3. Neutrino Regression Methods	26
6. ML-Based Reconstruction of Dileptonic Top-Quark Pair Events	28
6.1. Jet-Assignment	29
6.1.1. Neural Network Architectures	29
6.1.2. Training Procedure	29
6.1.3. Hyperparameter Optimisation	29
6.2. Combination with Neutrino Regression	29
6.2.1. Unified Model Architecture	29
6.2.2. Training Strategy	29
6.2.3. Performance Evaluation	29
6.2.4. Multitask Learning Benefits	29
7. Performance-Evaluation of the ML-Based Reconstruction	30
7.1. Assignment Accuracy	30
7.2. Variable Reconstruction	30
7.3. Region Specific Performance	30
8. Data-Driven Validation Studies	31
8.1. Run 2 Data Sample	31
8.2. Analysis Setup	31
8.3. Results	31
9. Conclusion	32
9.1. Summary	32
9.2. Outlook	32
A. Additional Plots	33
B. Supplementary Computations and Proofs	34
B.1. Mathematical Foundation of Machine Learning	34
B.1.1. Mean Squared Error Loss and Conditional Expectation	34

1. Introduction

The top quark, being the heaviest known elementary particle, plays a crucial role in the Standard Model of particle physics. Due to its large mass, it stands out in two ways; for one, as having the highest Yukawa-coupling to the Higgs boson and secondly, as being the shortest lived elementary particle. While, the Yukawa-coupling makes it very interesting for Higgs physics, the short lifetime of the top quark presents it as the only window allowing us to look at the bare coupling of quarks. The top quark decays almost exclusively into a W boson and a b -quark. These properties make processes involving top quarks as some of the most studied at the Large Hadron Collider (LHC) at CERN.

Top quarks are predominantly produced in pairs ($t\bar{t}$) via the strong interaction in proton-proton collisions at the LHC. Each W boson coming from the top decay can decay either leptonically (into a charged lepton and neutrino) or hadronically (into a pair of quarks). The dileptonic decay channel, where both W bosons decay leptonically, results in a final state with two charged leptons, two b -jets, and two neutrinos. This channel, while having a lower branching ratio compared to other decay modes, offers a cleaner experimental signature due to the presence of leptons and reduced hadronic activity. Leptons in particular can be reconstructed with a high efficiency in a hadron collider environment. However, the presence of two neutrinos in the final state poses significant challenges for event reconstruction, as they evade detection, leading to missing transverse energy ($\cancel{E}_T$) in the event.

Due to its clean signature and sensitivity to various top-quark properties, the dileptonic $t\bar{t}$ channel is widely used for precision measurements of the top properties and spin correlations in particular. Two recent highlights in top-quark physics are the tests of spin correlations and quantum entanglement in pairs of top quarks [1], and the observation of a possible quasi-bound state resonance in the $t\bar{t}$ invariant mass spectrum [2, 3]. Both effects are predominantly studied in the dilepton decay channel of the top quark in a mass range close to the on-shell production threshold of two top quarks $m(t\bar{t}) \gtrsim 2 \cdot m_t \approx 350$ GeV. Compared to hadronic decays, the leptons allow a more accurate probing of the top quarks spin by preserving a larger fraction of the spin information [4, 5].

Both analyses rely on measurements of angular distributions and correlations between the decay products of the top quarks. Probing this system requires a precise reconstruction of the top quarks from their decay products, which is complicated by the presence of the two neutrinos. While analytical neutrino regression strategies have been studied extensively, the problem of correctly assigning b -jets to their parent top quarks remains largely unstudied. In channels with hadronically decaying top quarks, various methods for jet assignment have been developed and employed successfully [6]. However, in the dileptonic channel, the jet assignment problem was often simplified or overlooked due to a focus on neutrino reconstruction. However, many of the sensitive variables used in measurements of spin correlations depend critically on the correct assignment of the jets.

1. Introduction

Misassignments degrade the resolution of reconstructed observables, ultimately limiting the precision of measurements.

2. Theoretical Background

2.1. The Standard Model of Particle Physics

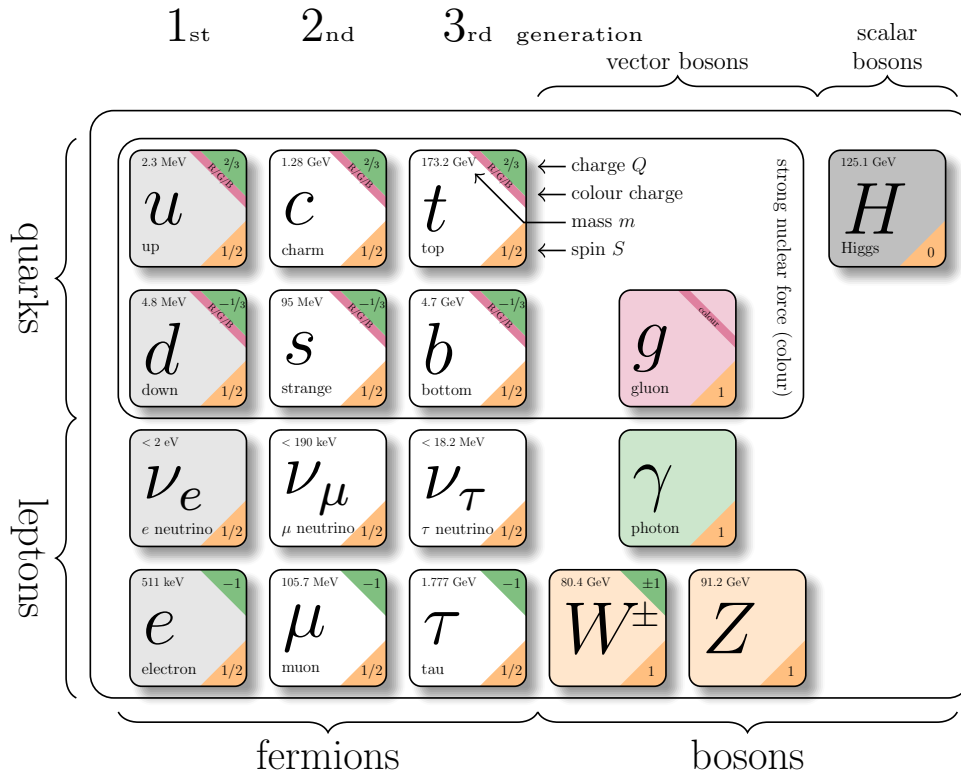


Figure 2.1.: The particles of the Standard Model. Particle properties taken from Reference [7].

The Standard Model (SM) of particle physics is a quantum field theory describing three of the four fundamental interactions of nature—electromagnetic, weak and strong—and the elementary particles on which they act. It is formulated as a renormalizable, Lorentz-invariant gauge theory with gauge group

$$SU(3)_C \times SU(2)_L \times U(1)_Y,$$

and its structure has been established in a series of seminal works [8–13]. A summary of the particle content is shown in Figure 2.1.

The elementary particles fall into two categories, fermions and bosons. Fermions are spin- $\frac{1}{2}$ fields that constitute matter, organised into three generations of quarks and leptons.

2. Theoretical Background

Quarks carry colour charge and participate in the strong interaction; leptons do not. Bosons are integer-spin gauge fields mediating the fundamental forces: the photon for electromagnetism, the $W^\pm$ and Z bosons for the weak interaction, and eight gluons for the strong interaction. The Higgs boson completes the SM and is responsible for electroweak symmetry breaking (EWSB), giving mass to the weak gauge bosons and, through Yukawa couplings, to the fermions. The SM Lagrangian can be written schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Yukawa}} + \mathcal{L}_{\text{Higgs}},$$

where the first term contains the gauge kinetic terms and self-interactions, the second describes the fermions and gauge covariant derivatives, the third encodes the Yukawa couplings to the Higgs field, and the last term contains the Higgs potential and EWSB mechanism.

Below, the three interaction sectors are briefly summarised, with emphasis on the aspects relevant for top-quark physics and LHC phenomenology.

2.1.1. Electromagnetic Interaction

The electromagnetic interaction is described by quantum electrodynamics (QED), the $U(1)_{\text{EM}}$ gauge theory that remains after EWSB. Its massless gauge boson, the photon, couples to electric charge and does not self-interact. Because QED is an Abelian theory, its coupling runs only mildly with energy and remains perturbative at all experimentally accessible scales.

2.1.2. Weak Interaction

The weak interaction forms, together with electromagnetism, the electroweak theory based on the gauge group $SU(2)_L \times U(1)_Y$. The vacuum expectation value of the Higgs field spontaneously breaks the symmetry to $U(1)_{\text{EM}}$, giving masses to the weak gauge bosons through the Higgs mechanism [14–16]. The resulting massive $W^\pm$ and Z bosons mediate a short-range force.

A defining property of the weak interaction is its *chiral* structure: charged-current interactions couple only to left-handed fermions and right-handed antifermions. This feature, predicted in [17] and confirmed in the Wu experiment [18], leads to maximal parity violation and characteristic angular distributions of weak decay products. The vertex factor of the charged-current interaction is given by

$$\frac{-ig}{2\sqrt{2}}\gamma^\mu(1 - \gamma^5), \tag{2.1}$$

where g is the weak coupling constant, γ^μ are the Dirac matrices. The projection operator $(1 - \gamma^5)$ leads to a coupling exclusively to the left-handed components. Based on this, the left-handed particles are arranged in $SU(2)_L$ doublets, while the right-handed particles are singlets.

For quarks, the weak eigenstates d', s', b' are not identical to their mass eigenstates d, s, b . This misalignment is described by the Cabibbo-Kobayashi-Maskawa (CKM) matrix [19,

2. Theoretical Background

20], which enables flavour-changing coupling in charged-current interactions.

2.1.3. Strong Interaction

The strong interaction is described by quantum chromodynamics (QCD), a non-Abelian gauge theory with gauge group $SU(3)_C$. Quarks carry colour charge, while gluons carry a colour and anticolour index, giving rise to self-interactions through three- and four-gluon vertices. These features lead to two fundamental properties:

Asymptotic freedom. The QCD coupling decreases at high momentum transfer [11, 21], allowing perturbative calculations of hard processes such as top-quark pair production. In the high-energy regime of the LHC, quarks and gluons effectively behave as quasi-free particles.

Confinement. At low energies, the QCD coupling becomes large, preventing coloured states from existing in isolation. Instead, they form colour-neutral bound states—hadrons. The transition from short-distance partons to long-distance hadrons involves nonperturbative dynamics encapsulated through hadronisation models.

A crucial consequence for collider physics is that quarks and gluons produced in hard interactions undergo a cascade of QCD emissions before hadronising into jets. This process is typically described in terms of:

- **Parton distribution functions (PDFs)**, characterising the proton structure and entering via the QCD factorisation theorem [22].
- **Initial- and final-state radiation (ISR/FSR)**, arising from soft and collinear gluon emission.
- **Parton showers**, which resum logarithmically enhanced emissions and are implemented in Monte Carlo generators such as PYTHIA, first developed in [23, 24] and extended in modern formulations [25].
- **Hadronisation models**, such as the Lund string model [26, 27], which describe the confinement-driven formation of hadrons.

These aspects of QCD are essential for the reconstruction of hadronic final states and for Monte Carlo simulations of LHC physics.

2.2. Top-Quark Physics

The top quark is the heaviest known elementary particle. The prediction of its existence arose due to the already observed CP-Violation in the kaon system, which required a third generation of quarks to be explained within the SM framework [20]. Furthermore, the discovery of the bottom quark in 1977 [28] implied the existence of its weak isospin partner, the top quark. The top quark was eventually discovered in 1995 by the CDF and DØ

2. Theoretical Background

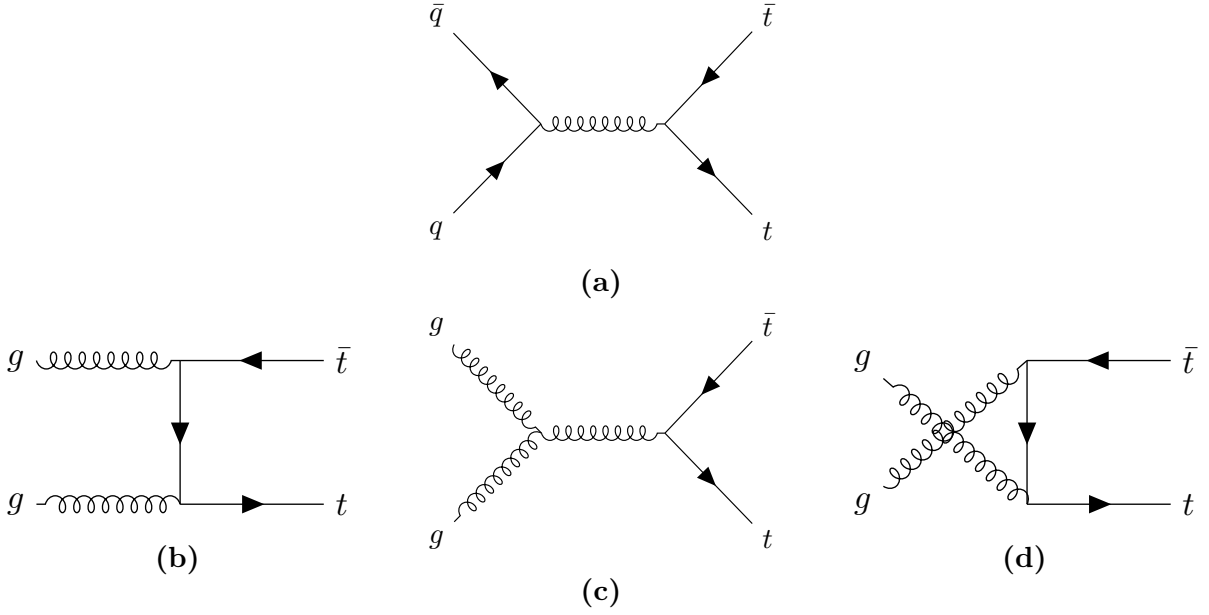


Figure 2.2.: Leading order Feynman diagrams for top-quark pair production in (a) quark-antiquark annihilation and (b), (c) & (d) gluon fusion (ggF).

collaborations at the TEVATRON collider [29, 30]. Currently, the world average of the top-quark mass is measured to be [7]

$$m_t = 172.56 \pm 0.31 \text{ GeV}. \quad (2.2)$$

Along with the largest mass, the top quark also has the strongest Yukawa coupling to the Higgs boson, making it interesting for studies of the Higgs mechanism and electroweak symmetry breaking.

Due to its large mass, the top quark also has the shortest lifetime of all quarks, about 5×10^{-25} s [7], which is shorter than the typical hadronisation timescale of 3×10^{-24} s [31]. Therefore, the top is the only quark that decays before it hadronises, allowing for direct studies of its properties through its decay products.

Notably, the top quark also decays before its spin can decorrelate, preserving spin information in the angular distributions of its decay products [32]. This unique feature enables precise measurements of spin correlations and polarisation of top quarks.

At hadron colliders like the LHC, top quarks are predominantly produced in pairs via the strong interaction. The leading order (LO) Feynman diagrams for top-quark pair production are shown in Figure 2.2. At the LHC, the dominant production mechanism is gluon fusion (ggF). This is due to the high gluon density in the proton at the relevant Bjorken- x values for top-quark production.

The CKM matrix element V_{tb} is close to unity [7], leading to a nearly exclusive decay of the top quark into a W boson and a b quark. The W can subsequently decay either leptonically into a charged lepton and a neutrino, or hadronically into a pair of quarks, with branching ratios of about $\text{BR}(W \rightarrow \ell \nu) = 1/3$ and $\text{BR}(W \rightarrow q \bar{q}) = 2/3$, respectively [7].

2. Theoretical Background

Based on these W decay modes, top-quark pair events are categorised into three channels. In the **dileptonic channel**, both W bosons decay leptonically ($W^+ \rightarrow \ell^+ \nu_\ell$, $W^- \rightarrow \ell^- \bar{\nu}_\ell$), resulting in two charged leptons, two neutrinos and two b quarks in the final state. This channel has a branching ratio of approximately $1/9$. The **semileptonic channel** occurs when one W boson decays leptonically and the other hadronically ($W \rightarrow q\bar{q}'$), yielding one charged lepton, one neutrino, four quarks (including two b quarks) in the final state. This is the most common channel, with a branching ratio of about $4/9$. Finally, in the **all-hadronic channel**, both W bosons decay hadronically, producing six quarks (including two b quarks) in the final state. This channel has the highest branching ratio of approximately $4/9$ but suffers from large multijet background contamination in the busy environment of hadron colliders like the LHC.

2.2.1. Spin Correlations in Top-Quark Pairs

As elaborated upon before, the top quark decays before it can hadronise or its spin decorrelates. Therefore, the spin information of the top quark is directly transferred to its decay products. In top-quark pair production, the spins of the top and antitop quarks are correlated due to the production mechanism. Due to the chiral nature of the weak interaction, these spin correlations manifest in the angular distributions of the decay products, particularly the charged leptons in the dileptonic decay channel.

Recent studies at the LHC began to explore potential effects of quantum entanglement in top-quark pairs [1]. Entanglement generally describes a quantum mechanical phenomenon where the quantum states of two or more particles become correlated in such a way that the state of one particle cannot be described independently of the state of the other(s), even when the particles are separated by large distances [33]. This property is expressed as the system having a non-separable density matrix. While a general description of this phenomenon is beyond the scope of this thesis, entanglement can be tested using specific criteria. One such criterion is the *Peres-Horodecki criterion* [34, 35], which states that if the partial transpose of the density matrix of a bipartite system has at least one negative eigenvalue, then the system is entangled. Considering the top-quark pair's spin as a bipartite system of two spin- $\frac{1}{2}$ particles (qubits), this criterion can be applied to test for entanglement in their spin states.

In Reference [36] it was proposed to measure entanglement in top-quark pairs produced at the LHC by analysing the angular distributions of their decay products, specifically the charged leptons in the dileptonic decay channel. The study showed that by measuring the opening angle between the two leptons, one can construct an observable sensitive to the entanglement of the spin system of the top quarks. Due to the kinematics of the top-quark pair production, the entanglement is expected to be more pronounced in certain regions of phase space, one such region is at low invariant masses of the top-quark pair system. One particular region of interest is the so-called *threshold region*, where the top-quark pair is produced with low relative velocity. In this region, the top quark is produced as a spin singlet state in about $\sim 80\%$ of the cases, leading to a strong entanglement signature [37]. The ATLAS collaboration has recently reported the first experimental evidence of quantum correlations consistent with entangled spin quantum states in top-quark pairs [1]. Measurements of this variable, are particularly sensitive in the dilepton channel, because,

2. Theoretical Background

the lepton has the highest spin-analysing power [4, 5]. The spin-analysing power measure, to which extent the spin-information of the parent is imprinted in the angular distributions of the decay products. Due to QCD effects, this is lower for light jets compared to leptons.

2.2.2. Top-Quark Pair Qausi-Bound-State Effects

Even though the top quark decays before it can hadronise, near the production threshold of top-quark pairs, the strong interaction between the top and top antiquarks can lead to the formation of transient bound states, coined as toponium [37–39]. These bound state effects can influence the production cross section and kinematic distributions of top-quark pairs near threshold. The CMS collaboration has recently observed an excess in the invariant mass distribution of top-quark pairs near threshold, consistent with the presence of toponium bound state effects [3]. The modelling of these effects is based on non-relativistic QCD (NRQCD) calculations, which account for the strong interaction dynamics between the top and top antiquarks in the near-threshold region [40, 41]. Understanding and accurately modelling these bound state effects is crucial for precision measurements of top-quark properties and for searches for new physics in top-quark pair production.

This study is also investigating the threshold region of top-quark pair production and makes use of angular correlations between the decay products as a probe of the underlying dynamics, including potential bound state effects.

2.2.3. Dileptonic Decay Channel

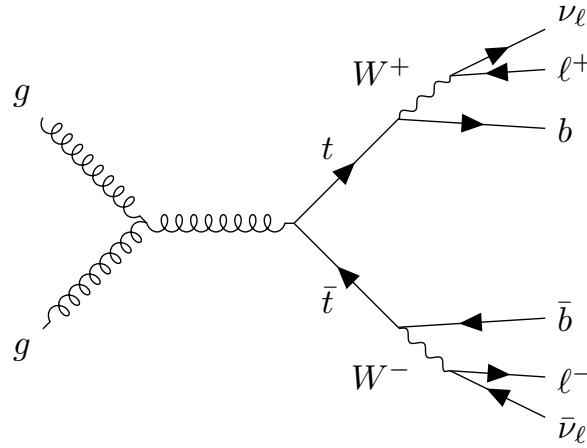


Figure 2.3.: Leading order Feynman diagram for top-quark pair production with dileptonic decay.

The dileptonic decay channel, where both W bosons decay leptonically, has a branching ratio of about $1/9$ and results in a final state with two charged leptons, two neutrinos and two b quarks, as illustrated in Figure 2.3. Due to the lower branching ratio, the dileptonic channel has a smaller event yield compared to the other channels. However, it offers a cleaner signature with reduced background contamination, making it particularly suitable

2. Theoretical Background

for precision measurements of top-quark properties. The presence of two neutrinos in the final state leads to missing transverse energy ($\cancel{E}_T$) in the event, complicating the full reconstruction of the top quark's kinematics. Advanced techniques, such as kinematic fitting and multivariate analyses, are often employed to address these challenges and extract maximum information from the dileptonic events.

Kinematic Constraints

Assuming both W bosons and top quarks are on-shell, the following kinematic constraints can be applied to the dileptonic decay channel,

$$(p_{\ell_1} + p_{\nu_1})^2 = m_W^2, \quad (2.3)$$

$$(p_{\ell_2} + p_{\nu_2})^2 = m_W^2, \quad (2.4)$$

$$(p_{b_1} + p_{\ell_1} + p_{\nu_1})^2 = m_t^2, \quad (2.5)$$

$$(p_{b_2} + p_{\ell_2} + p_{\nu_2})^2 = m_t^2, \quad (2.6)$$

$$(\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_x = \cancel{E}_{Tx}, \quad (2.7)$$

$$(\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_y = \cancel{E}_{Ty}. \quad (2.8)$$

If one assumes the neutrinos to be massless, these six equations provide constraints on the six unknown components of the two neutrino momenta. Due to the algebraic nature of the constraints, multiple solutions can exist for a given event, leading to ambiguities in the reconstruction. Due to the effects of detector resolution and off-shell particles, this system of equation can also have no real-valued solution [42, 43]. Various methods have been developed to address these challenges, including likelihood-based approaches [44], matrix element methods, and machine learning techniques [45], which aim to optimally utilise the available information and improve the accuracy of the top quark's kinematic reconstruction in dileptonic events.

2.2.4. Spin-Sensitive Angular Variables

To study spin correlations and entanglement in top-quark pairs, a spin-sensitive angular variable is used. For the measurements of the quasi-bound-state effects, an additional angular variable is employed. Both variables are defined using the charged leptons from the dileptonic decay of the top-quark pair.

To define the variables, the momenta of the leptons are first boosted into their respective parent top-quark rest frames $\hat{\ell}_t^+$ and $\hat{\ell}_t^-$. The inner product of these unit vectors then defines the variable

$$c_{\text{hel}} = \hat{\ell}_t^+ \cdot \hat{\ell}_t^-. \quad (2.9)$$

Analogously, one can define the variable c_{han} , where either of the lepton momenta components parallel to the parent top-quark momentum in the $t\bar{t}$ rest frame is flipped before taking the inner product [46, 47]. This variable adds important context by enabling isolation of scalar states.

potentially
update
cita-
tion
from
preprint

2. Theoretical Background

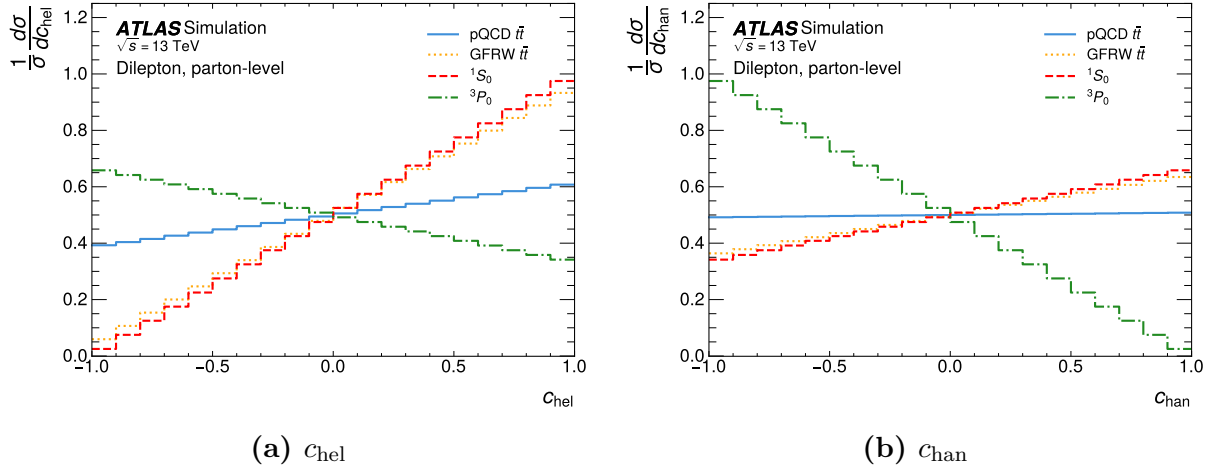


Figure 2.4.: Simulation of the distributions of c_{hel} and c_{han} for dileptonic decaying top-quark pairs produced near threshold for different spin states and combinations thereof. The distributions are normalised to unity and at parton level. The Figures are taken from Reference [2].

3. Machine Learning in Particle Physics

Since this work focuses on improving the event reconstruction of dileptonic $t\bar{t}$ decays using machine learning techniques, this chapter aims to provide an overview over the fundamental concepts and methodologies employed in machine learning.

From a computational perspective, machine learning can be viewed as the optimization of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that maps input data points $\mathbf{x} \in \mathbb{R}^n$ to output predictions $\mathbf{y} \in \mathbb{R}^m$. The function f is parameterised by a set of learnable parameters θ , which are adjusted during the training process to minimise a predefined *loss function* $L(\mathbf{y}, f(\mathbf{x}; \theta))$. The loss function quantifies the discrepancy between the predicted outputs and the true target values, guiding the optimization process.

3.1. Supervised Learning

Machine learning can be broadly categorised into supervised and unsupervised learning. In supervised learning, the model is trained on a labelled dataset, where each input data point is associated with a corresponding target output. The goal of the model is to learn a mapping from inputs to outputs, enabling it to make accurate predictions on unseen data.

3.1.1. Monte Carlo Training Data

Events are usually simulated using a cascade of different programs, each simulating a different state of the event. First the hard scattering process is simulated using matrix element generators. These programs calculate the probabilities of different particle interactions based on the underlying physics theories, such as the Standard Model. Using these probabilities, they generate events that represent the initial state of the particles after the collision. The possible kinematic phase space is sampled according to these probabilities, resulting in a set of particles with specific momenta and energies. This type of event variables is often referred to as *parton-level*.

Next, the parton showering and hadronization processes are simulated. Parton showering models the emission of additional particles from the initial partons, while hadronization simulates the formation of hadrons from quarks and gluons. These processes are crucial for accurately modelling the final state particles observed in detectors.

Finally, detector simulation programs are used to simulate the interaction of particles with the detector material, producing realistic detector responses. This includes simulating the energy deposits in calorimeters, hits in tracking detectors, and other relevant signals.

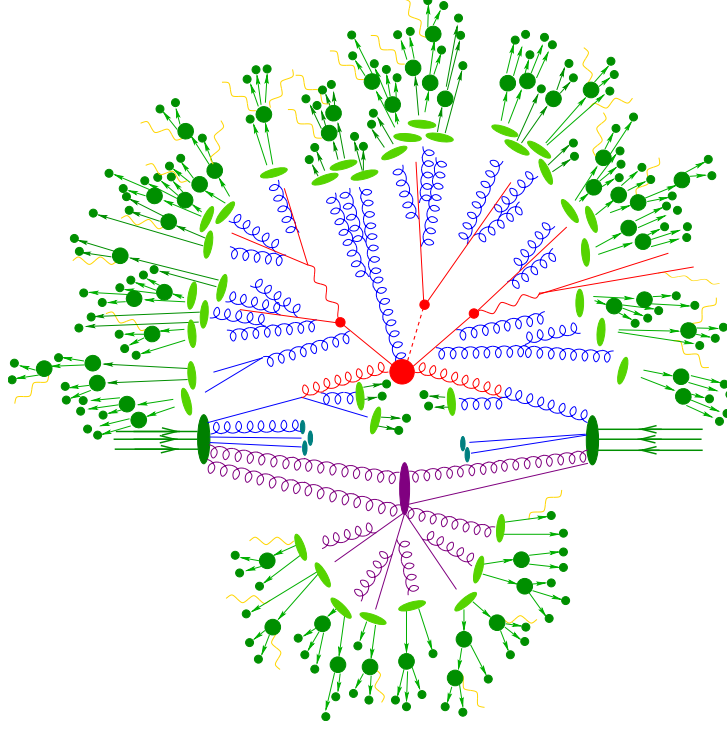


Figure 3.1.: Schematic representation of the Monte Carlo event generation process in particle physics. Colouring of vertices and particles indicates the event generation stage: hard scattering process (big red blob), QCD radiation (blue), hadronisation (light green), showering (dark green), and a secondary vertex (purple). The Figure is taken from [48].

While the actual detector response is simulated using complex detector simulation software, for many machine learning applications, a simplified representation of the detector response is sufficient. This can involve smearing the particle momenta and energies according to the detector resolution, applying efficiency corrections, and simulating the effects of pile-up.

This is because, for the reconstruction of the physics objects, such as jets, leptons, and missing transverse energy, highly sophisticated algorithms are used that already take into account the detector effects (Note that these algorithms may also be machine learning based). Therefore, the input features for machine learning models can often be derived directly from these reconstructed objects, rather than relying on the raw detector signals. The reconstructed object event variables are called *reco-level*.

3.1.2. Training

During the training phase, the model is presented with a set of input features derived from the reco-level event variables, along with their corresponding target outputs, which are typically derived from the parton-level event variables. The model learns to map the input features to the target outputs by minimizing a loss function that quantifies the difference between the predicted and true values. Common loss functions include mean squared

error for regression tasks and cross-entropy loss [49] for classification tasks. The training process involves iteratively (each step is called *epoch*) updating the model's parameters using an optimization algorithm to minimise the loss function.

3.1.3. Validation and Testing

To evaluate the performance of the trained model, it is essential to validate and test it on independent datasets that were not used during training. This helps to assess the model's generalization capabilities and ensures that it can make accurate predictions on unseen data. A common practice is to split the available data into training, validation, and test sets. The validation set is used to tune hyperparameters and monitor the model's performance during training, while the test set is reserved for the final evaluation of the model's performance.

Since in particle physics, simulated data is often used for training and background modelling, one typically employs a k-folding strategy [50], where the data is divided into k subsets. The model is trained k times, each time using a different subset as the validation set and the remaining subsets for training. This approach helps to use the available data more efficiently.

3.2. Regression and Conditional Density Learning

To highlight some important distinctions between different machine learning tasks, it is useful to briefly discuss regression and conditional density learning.

3.2.1. Supervised Learning as Risk Minimisation

In supervised learning, the goal is to learn a parametrised function f_θ that maps input data $\mathbf{x}$ to output predictions $\mathbf{y}$. This can be formalised as a risk minimisation problem, where the objective is to minimise the expected loss over the joint distribution of inputs and outputs. The expected risk can be expressed as:

$$R(f) = \mathbb{E}_{\mathbf{x}, \mathbf{y}}[L(\mathbf{y}, f_\theta(\mathbf{x}))] \quad (3.1)$$

where L is the loss function that quantifies the discrepancy between the true output $\mathbf{y}$ and the predicted output $f_\theta(\mathbf{x})$. The goal of training is to find the parameters θ that minimise this expected risk, which is typically approximated using empirical risk minimisation on the training dataset.

3.2.2. Regression

In regression tasks, the output $\mathbf{y}$ is a continuous variable, and the model is trained to predict a specific value for each input $\mathbf{x}$. The loss function commonly used for regression is the mean squared error (MSE), which measures the average squared difference between

the predicted values and the true values:

$$L_{\text{MSE}}(\mathbf{y}, f(\mathbf{x})) = \frac{1}{N} \sum_{i=1}^N (y_i - f_{\theta}(\mathbf{x}_i))^2 \quad (3.2)$$

where N is the number of samples in the dataset. The model learns to minimise this loss function. If the labels are not deterministic, meaning that there is inherent uncertainty in the target values, one can describe the labels using a conditional probability distribution $p(\mathbf{y}|\mathbf{x})$. In this case, minimising the MSE corresponds to learning the conditional mean of the distribution (see Chapter B for a proof):

$$f_{\theta}(\mathbf{x}) = \mathbb{E}[\mathbf{y}|\mathbf{x}] = \int \mathbf{y} p(\mathbf{y}|\mathbf{x}) d\mathbf{y}. \quad (3.3)$$

Depending on the loss-function used, the model can learn different statistics of the conditional distribution. For example, using the mean absolute error (MAE) loss function would lead to learning the conditional median instead of the mean.

3.2.3. Conditional Density Learning

In contrast to regression, conditional density learning aims to learn the entire conditional probability distribution $p(\mathbf{y}|\mathbf{x})$ rather than just a point estimate. This is particularly useful when the target variable has inherent uncertainty or when multiple outcomes are possible for a given input. To quantify the difference between the true conditional distribution and the model's predicted distribution, the Kullback-Leibler (KL) divergence is often used as a loss function [51]

$$L_{\text{KL}}(p, q) = \int p(\mathbf{y}|\mathbf{x}) \log \left(\frac{p(\mathbf{y}|\mathbf{x})}{q_{\theta}(\mathbf{y}|\mathbf{x})} \right) d\mathbf{y} \quad (3.4)$$

where $p(\mathbf{y}|\mathbf{x})$ is the true conditional distribution and $q_{\theta}(\mathbf{y}|\mathbf{x})$ is the model's predicted distribution. Minimizing this loss function encourages the model to produce a predicted distribution that closely matches the true distribution, allowing for a more comprehensive understanding of the uncertainty and variability in the target variable. In practice, computing KL divergence can be challenging, since the true distribution might be unknown or difficult to estimate. In such cases, one relies on the data to provide an empirical estimate of the true distribution, and the loss function is approximated using samples from the dataset. It can be shown that minimizing the KL divergence is equivalent to maximizing the likelihood of the observed data under the model's predicted distribution, which is a common approach in training probabilistic models. For the case of a discrete target variable, this corresponds to minimising the cross-entropy loss, which is widely used in classification tasks [49].

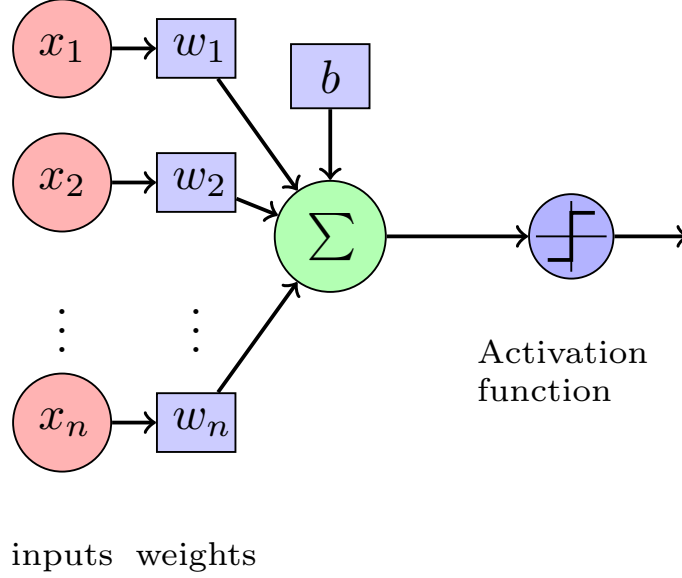


Figure 3.2.: Schematic of a single artificial neuron (perceptron). The neuron receives multiple input signals, each associated with a weight, and computes a weighted sum of these inputs, optionally a so-called bias is added. An activation function is then applied to this sum to produce the neuron’s output.

3.3. Neural Networks

As elaborated before, machine learning models can be viewed as parameterised functions that map input data to output predictions. Neural networks are a class of machine learning models that are inspired by the structure and function of biological neural networks in the human brain. They consist of interconnected artificial neurons that process and transform the input data to produce the desired output. The simplest unit of a neural network is the artificial neuron, also known as a perceptron [52]. A perceptron takes multiple input signals, each associated with a weight, and computes a weighted sum of these inputs. An activation function is then applied to this sum to produce the neuron’s output. This structure is illustrated in Figure 3.2. The general structure of a perceptron is inspired by biological neurons. The activation function introduces non-linearity into the model, allowing it to learn complex patterns in the data. Common activation functions include the sigmoid function, hyperbolic tangent ($\tanh$), and rectified linear unit (ReLU) [53]. For this work, the ReLU activation function is used, which is defined as $f(x) = \max(0, x)$, meaning that it outputs the input directly if it is positive, and zero otherwise. This activation function has been shown to be effective in training deep neural networks by mitigating the vanishing gradient problem [53].

3.3.1. Dense Neural Network

A dense neural network (DNN), also known as a fully connected neural network [54], is a type of artificial neural network where each neuron in one layer is connected to every

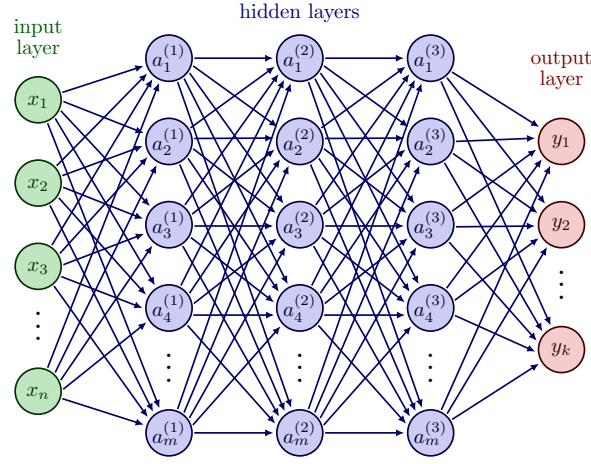


Figure 3.3.: Schematic of a dense neural network with multiple layers. Each layer consists of multiple neurons, and each neuron in one layer is connected to every neuron in the subsequent layer.

neuron in the subsequent layer. This architecture allows for the learning of complex relationships between input features and output targets. A schematic of a dense neural network is shown in Figure 3.3.

A DNN consists of an input layer, one or more hidden layers, and an output layer. The input layer receives the input features, while the hidden layers perform a series of transformations on the data using the weights and biases associated with each neuron. The output layer produces the final predictions. Each layer applies an activation function to the weighted sum of inputs from the previous layer, enabling the network to learn non-linear mappings. The depth (number of layers) and width (number of neurons per layer) of the network can be adjusted.

3.3.2. Recurrent Neural Networks

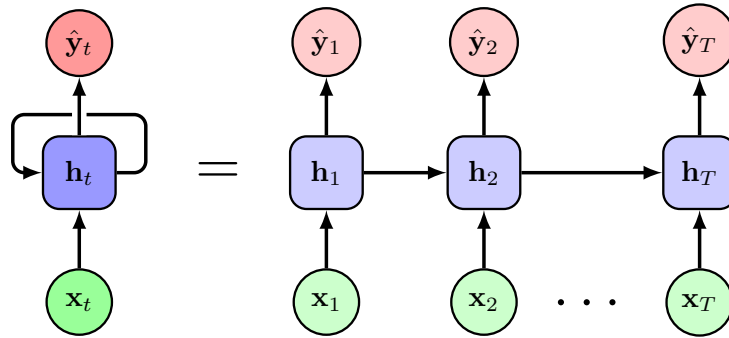


Figure 3.4.: Schematic of a recurrent neural network (RNN). The RNN processes sequences of data by maintaining a hidden state that captures information from previous time steps. The unfolded representation illustrates how the RNN operates over multiple time steps.

Dense neural networks require their inputs to have a fixed size. However, in many applications, including particle physics, the input data can vary in size. For example, the number of particles detected in an event can differ from one event to another. To handle such sequential data, recurrent neural networks (RNNs) [55] are employed. RNNs are designed to process sequences of data by maintaining a hidden state that captures information from previous time steps. A schematic of an RNN is shown in Figure 3.4. RNNs consist of a series of interconnected neurons that process input data sequentially. At each time step, the RNN takes an input vector and combines it with the hidden state from the previous time step to produce a new hidden state. This hidden state is then used to generate the output for the current time step. The recurrent connections allow the RNN to retain information from previous inputs, enabling it to learn temporal dependencies in the data. One common variant of RNNs is the Long Short-Term Memory (LSTM) network [56], which addresses the vanishing gradient problem and allows for learning long-term dependencies more effectively. Another caveat of RNNs is that they are inherently sequential, making them less suited for data without inherent order, such as sets of particles in an event.

3.3.3. Transformer Networks

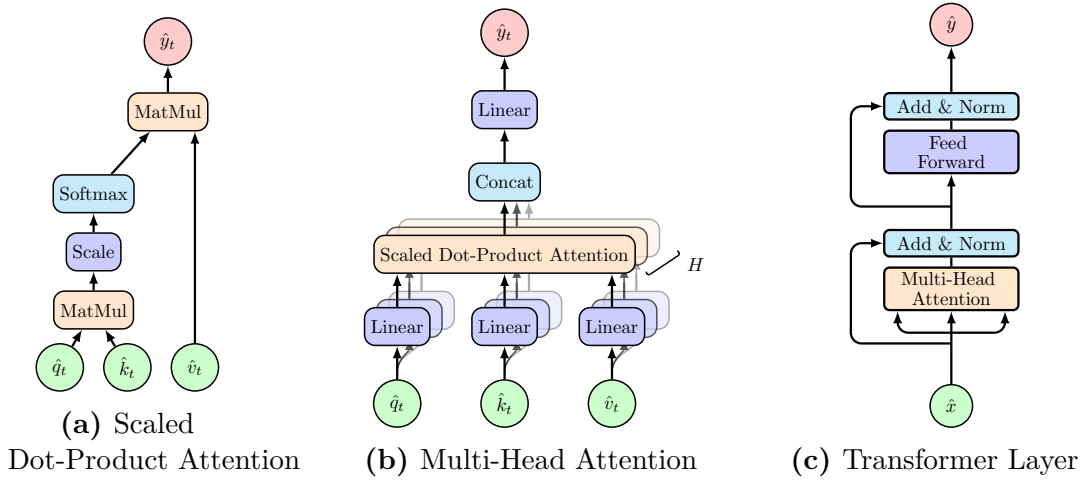


Figure 3.5.: Different components of the transformer architecture from smallest to largest unit: (a) Scaled Dot-Product Attention, (b) Multi-Head Attention, and (c) Transformer Layer.

The transformer architecture deals with sequential data, similar to RNNs, but it does so using an attention mechanism rather than recurrent connections. This allows transformers to process information in parallel, making them more efficient for training on large datasets. Additionally, transformer can capture long-range dependencies in the data more effectively than RNNs, as they relate all elements of the input sequence to each other through attention mechanisms.

The core component of the transformer architecture is the attention mechanism, which allows the model to focus on different parts of the input sequence when making predictions.

3. Machine Learning in Particle Physics

The most commonly used attention mechanism is the scaled dot-product attention [57], illustrated in Figure 3.5a. In this mechanism, three vectors are computed for each input element: the query (q), key (k), and value (v) vectors. The attention scores are calculated by taking the dot product of the query and key vectors, scaling them by the square root of the dimension of the key vectors, and applying a softmax function to obtain attention weights. These weights are then used to compute a weighted sum of the value vectors, producing the output of the attention mechanism. Multi-head attention, shown in Figure 3.5b, extends the scaled dot-product attention by allowing the model to attend to different parts of the input sequence simultaneously. This is achieved by using multiple parallel attention heads, each with its own set of learned linear transformations for the query, key, and value vectors. The outputs of all attention heads are concatenated and linearly transformed to produce the final output.

The key, query and value vectors are typically obtained by applying learned linear transformations to the input embeddings. Depending on the type of relational information one wants to capture, one can choose to derive these vectors from different input sources. For example, in the *self-attention mechanism*, all three vectors are derived from the same input sequence ($\hat{q}_t = \hat{k}_t = \hat{v}_t = \hat{x}_t$), allowing the model to relate different elements within the same sequence. In *cross-attention*, the query vectors are derived from one sequence, while the key and value vectors are derived from another sequence ($\hat{q}_t = \hat{x}_{1,t}$, $\hat{k}_t = \hat{v}_t = \hat{x}_{2,t}$), enabling the model to relate information between two different sequences.

Symmetries In many applications, including particle physics, the input data exhibits certain symmetries that can be exploited to improve model performance.

The attention mechanism has inherent permutation equivariances, that can be exploited. Self-attention is permutation equivariant to the input sequence, meaning that permuting the input elements results in a corresponding permutation of the output elements. This property is particularly useful when dealing with sets or unordered data, where the order of the elements does not carry any inherent meaning. Cross-attention is permutation equivariant to permutations of the query inputs and separately to permutations of the key/value inputs. This allows the model to effectively relate information between two different sequences, regardless of their order.

3.3.4. Training Neural Networks

Training neural networks involves optimizing the model's parameters (weights and biases) to minimise the loss function. The number of parameters in a neural network can rise rapidly, especially in deep architectures, making the optimization process computationally intensive. The most commonly used optimization algorithm for training neural networks is stochastic gradient descent (SGD) [58], which iteratively updates the model's parameters based on the gradients of the loss function with respect to the parameters. Modern variants of SGD, such as Adam [59], incorporate adaptive learning rates and momentum to improve convergence. The backpropagation algorithm [54] is used to efficiently compute the gradients of the loss function with respect to the model's parameters. It involves propagating the error signal backward through the network, allowing for the calculation of gradients at each layer. These gradients are then used by the optimization algorithm to

3. Machine Learning in Particle Physics

update the parameters. The algorithm relies on the chain rule and nested structure of the neural network to compute the gradients efficiently. Regularization techniques, such as dropout [60] and weight decay, are often employed during training to prevent overfitting and improve the model's generalization capabilities. Dropout randomly deactivates a fraction of the neurons during training, forcing the network to learn more robust features. Weight decay [61] adds a penalty term to the loss function based on the magnitude of the weights, discouraging overly complex models and promoting generalization.

4. Experimental Setup

4.1. The LHC

The Large Hadron Collider (LHC) at CERN is a synchrotron designed to accelerate protons and lead nuclei. This work focuses on the proton-proton collisions during Run 2. During Run 2 from 2015 to 2018, the LHC operated at a centre of mass energy of $\sqrt{s} = 13 \text{ TeV}$ with an integrated luminosity of $\mathcal{L}_{int} \approx 140 \text{ fb}^{-1}$ [62].

After completion of Run 3 in June 2026 and an operational pause, the LHC will operate with increased luminosity as the High-Luminosity Large Hadron Collider (HL-LHC). During this phase, it is planned to accumulate data with an integrated luminosity of around $\mathcal{L} = 3 \text{ ab}^{-1}$ [63].

4.2. The Detector

The ATLAS detector is a general-purpose particle detector at the LHC at CERN. It can roughly be separated into three elements: the inner detector, the calorimeter and the muon spectrometer [64].

4.2.1. The Coordinate System

The interaction point marks the origin of the ATLAS coordinate system. The z -axis is orientated along the beamline, the x -axis points towards the centre of the LHC and the y -axis points upwards.

Using this convention, several kinematic variables are defined. One of which is the transverse momentum defined as

$$p_T = \sqrt{p_x^2 + p_y^2} \quad (4.1)$$

Because of the cylindrical shape of the ATLAS detector and the cylindrical symmetry of the interactions, the azimuthal angle is used as another variable. The polar angle relative to the z -axis can be used to define the pseudorapidity which can also be expressed in terms of the momentum

$$\eta = -\ln \left(\tan \left(\frac{\theta}{2} \right) \right) = \ln \left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z} \right). \quad (4.2)$$

In the ultra-relativistic limit $m \ll |\vec{p}|$ the pseudorapidity is equal to the rapidity defined

4. Experimental Setup

as

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right). \quad (4.3)$$

These variables are convenient for the usage in the ATLAS experiment because p_T , ϕ , and differences of η are invariant under Lorentz-boost along the z -axis.

4.2.2. Inner Detector

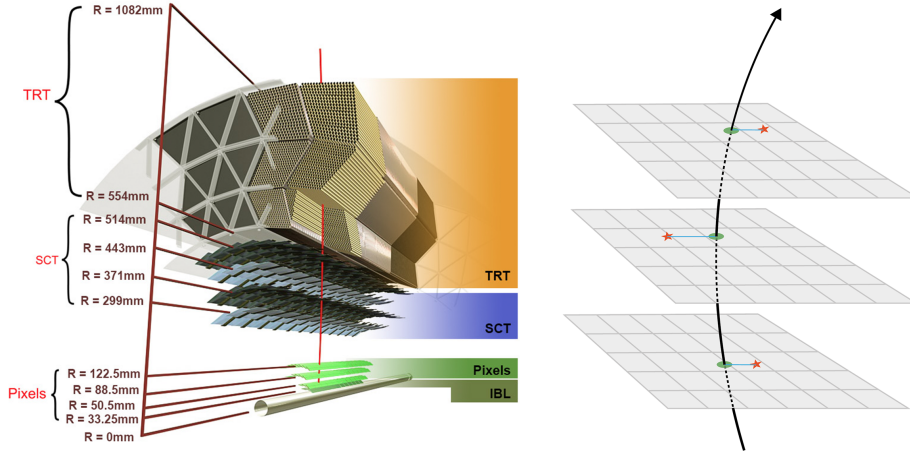


Figure 4.1.: Schematic cross-section of the inner detector of the ATLAS detector (© CERN).

The inner detector (ID) is the innermost part of the ATLAS detector. It is used for tracking charged particles, particle identification, and primary and secondary vertex determination [64]. The ID consists of three different sub-detectors. Their rough structure is depicted in Figure 4.1. The ID is encapsulated with a solenoid magnet providing a 2 T axial magnetic field on the inside of the ID. The magnetic field is crucial to measure the momentum of a charged particle. The transverse momentum is measured as the curvature of the particle's track due to the Lorentz force.

The detector part closest to the beam pipe is the pixel detector. It consists of 1744 modules arranged in three barrel layers. Each module hosts 47232 silicon pixels with a size of $50 \times 400 \mu\text{m}^2$. The pixels are semiconductor trackers used to detect the traversing of charged particles.

The following part of the detector is the semiconductor tracker (SCT). It consists of 4088 modules arranged in four layers to guarantee four position measurements of charged particles. Each module consists of four silicon sensors. The sensors offer a $17 \mu\text{m}$ resolution in-plane lateral and $580 \mu\text{m}$ in-plane longitudinal.

The outermost part of the inner detector is the transition radiation tracker (TRT). It consists of polyimide drift (straw) tubes with a 4 mm diameter that are arranged in a 528 mm thick cylindrical layer around the beam pipe. The straw tubes are interleaved with fibres for the readout. The transition radiation tracker utilises the transition light

4. Experimental Setup

emitted by charged particles traversing the interface between two media with different indices of refraction. The TRT offers a measurement of charged particles and electron identification.

4.2.3. Calorimeter System

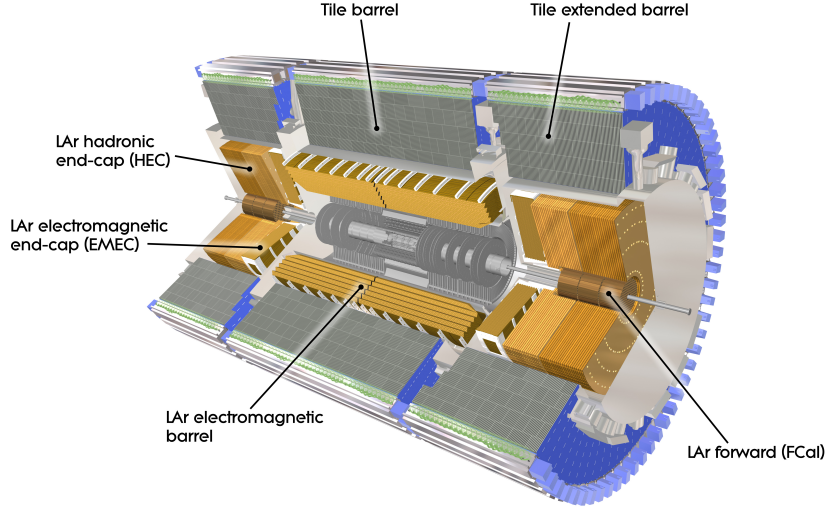


Figure 4.2.: Computer generated image of the ATLAS calorimeter (© CERN).

The calorimeters are the detector layers following the inner detectors. Its structure is divided into the Electromagnetic calorimeter (ECal) and the hadronic calorimeter (HCal).

Electromagnetic Calorimeter

The ECal is used for the energy and position measurement of electric charged particles and photons. It utilises bremsstrahlung and pair production to create a cascade of charged particles which are measured. It offers full azimuthal coverage and is equipped with end caps in the longitudinal direction of the beam pipe. The ATLAS ECal is a sampling calorimeter operating with lead as the passive and liquid Argon as the active medium. The innermost part of the ECal is a presampler which detects if the particle started showering before reaching the ECal.

Hadronic Calorimeter

The HCal measures the energy and position of baryons and mesons through strong interactions with the nuclei. In the range of $|\eta| < 1.7$ it is a sampling calorimeter with steel as the passive and scintillators as the active medium. For the end cap, liquid Argon is deployed as the active medium. The HCal works with the same principle as the ECal but offers less precise measurements.

In the range of $3.1 < |\eta| < 4.9$ ECal and HCal are substituted with the forward calorimeters (FCal) which are made up of three modules to fulfil the function of both calorimeters.

4. Experimental Setup

In combination with the FCal a total range of $|\eta| < 4.9$ is covered by the calorimeters. A computer-generated image of the structure of the different calorimeters used in the ATLAS detector is shown in Figure 4.2.

4.2.4. Muon Spectrometer

The muon spectrometer is the outermost part of the ATLAS detector. Its purpose is to detect charged particles exiting the calorimeters and measure their momentum in the range of $|\eta| < 2.7$. A detector dedicated to the measurement of muons is necessary because their mass makes them minimal ionizing particles in the energy range of the LHC collisions. The amount of energy loss due to bremsstrahlung is not sufficient to develop showers necessary to measure their energy. The muons' transverse momenta can be measured in the ID. Like the inner detector, the muon spectrometer utilises a magnetic field to conduct a measurement of the particle's momentum. In the muon spectrometer, however, a solenoid magnet is used to allow for the measurement of the muon's momentum along a different direction. Combining the measurements, one obtains full knowledge of the muons four-momentum. Further, does the high rate of stopped electrons and hadrons in the calorimeters enable a high specificity in the muon detection.

4.2.5. Trigger System

With a bunch spacing of 25 ns [62] collisions happen at a rate of 40 MHz. Each collision involves up to hundreds of particles. To reduce the data to a feasible amount, triggers are employed to filter less interesting events. Different layers of triggers operate either at the hardware or software level. The L1 trigger is a hardware level trigger and acts as the first filter for the events. It makes decisions in less than 2.5 μ s. The subsequent L2 trigger is a software level trigger and makes decisions in less than 200 μ s. Combined, the trigger system reduces the event rate from 40 MHz to 1000 Hz which are then stored at the CERN data centre.

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events

5.1. Data and Simulation

This section describes the data and simulation samples used in this analysis. At the current stage of this thesis, only simulation samples have been studied; the data samples will be described in the final version. The simulated samples are produced using Monte Carlo (MC) event generators as described in Section 3.1.1.

While a complete analysis requires detailed investigation of background processes contributing to the signal regions, this thesis focuses on developing and evaluating reconstruction methods. Therefore, only the signal process of $t\bar{t}$ production with dileptonic decays is described here. The background processes and their simulation will be discussed in the final thesis.

5.1.1. Simulation of Dileptonic Top-Quark Pair Events

The signal process of $t\bar{t}$ production with dileptonic decays is simulated using POWHEG [65] at next-to-leading order (NLO) in perturbative quantum chromodynamics (QCD). The POWHEG generator is interfaced with PYTHIA [25] for parton showering, hadronization, and underlying event modelling.

To link the reco-level particles to the parton-level objects, a parton matching is performed. The parton-level objects are defined as the top quarks and their decay products after final state radiation before hadronization. The matching is done by iteratively associating reconstructed jets to partons based on angular distance criteria. The decay products of the top quarks are matched within a cone of $\Delta R \leq 0.3$ and $\Delta R \leq 0.1$ for jets and leptons (e, μ) respectively. If multiple reconstructed objects are found within this cone, the one with the smallest ΔR is chosen.

5.1.2. Object Definition

The reconstruction of physics objects—electrons, muons, jets, and missing transverse energy ($\cancel{E}_T$)—is performed using the standard ATLAS reconstruction algorithms [64].

Electrons are required to have transverse momentum $p_T > 5$ GeV and pseudorapidity $|\eta| < 2.47$, excluding the transition region between the barrel and endcap calorimeters ($1.37 < |\eta| < 1.52$). Electrons must satisfy the Tight identification criteria and be isolated from other activity in the detector.

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events

Muons are reconstructed using information from both the inner detector and the muon spectrometer. They are required to have $p_T > 5$ GeV and $|\eta| < 2.5$. Muons must satisfy the Tight identification criteria and be isolated.

Jets are reconstructed using the anti- k_t algorithm [66] with a radius parameter of $R = 0.4$. They are required to have $p_T > 25$ GeV and $|\eta| < 2.5$. b -tagging is performed using the GN2 algorithm [67] to identify jets originating from b -quarks.

Missing Transverse Energy ($\cancel{E}_T$) is calculated from the negative vector sum of the transverse momenta of all reconstructed objects in the event, including a soft term to account for energy not associated with reconstructed objects.

5.1.3. Event Selection

Events are selected to contain exactly two oppositely charged leptons (electrons or muons). One of the leptons must have $p_T > 25$ GeV, while the other must have $p_T > 5$ GeV. At least two jets are required, with at least one being b -tagged at the 77% working point (WP). Additional selection criteria are applied to suppress background contributions, which will be detailed in the final thesis.

For the training and evaluation of the reconstruction methods, only events that pass the selection criteria and have a successful parton matching are considered.

5.2. Reconstruction of Top-Quark Pairs

Reconstructing the kinematics of dileptonic $t\bar{t}$ events requires an assignment of reconstructed objects to the parton-level decay products. This section describes the different reconstruction methods developed and evaluated in this thesis.

While the leptons are directly identified from the reconstructed objects, the assignment of jets to the b -quarks from the top quark decays is ambiguous due to the presence of multiple jets in the event. Additionally, the two neutrinos from the W boson decays are not directly detected, leading to missing information in the event reconstruction. Based on this, the reconstruction breaks down into two main tasks:

Jet Assignment Assigning the reconstructed jets to the b -quarks from the top quark decays.

Neutrino Regression Estimating the momenta of the two neutrinos using the measured $\cancel{E}_T$ and other event kinematics.

5.2.1. B-Tagging

The identification of b -jets is crucial for the reconstruction of $t\bar{t}$ events. Due to the long lifetime of b -hadrons, they travel a measurable distance from the primary vertex before decaying, leading to displaced vertices and tracks with large impact parameters. The GN2

algorithm [67] is used for b -tagging in this analysis, which combines various discriminating variables using a deep (graph) neural network to achieve high efficiency and purity in identifying b -jets. The performance of the b -tagging algorithm is characterised by its efficiency (the probability of correctly identifying a b -jet) and its misidentification rate (the probability of incorrectly tagging a non- b -jet as a b -jet).

5.2.2. Jet Assignment Methods

This thesis explores several methods for jet assignment, ranging from traditional algorithms to machine learning approaches. From recent publications e.g. [1], two conventional methods have been implemented as baselines:

Jet-Selection Both methods start by selecting the two b -jet candidates in the event based on the b -tagging at the 77% WP. If more than two b -tagged jets are present, the two with the highest p_T are chosen. If only one b -tagged jet is found, the non-tagged jet with the highest b -tagging score is selected as the second candidate.

χ^2 -Method

For both possible assignments of the two b -jet candidates to the parton-level b -quarks, the invariant masses of the reconstructed top quarks are computed,

$$\chi^2 = (m_{b_1, \nu, \ell^+} - m_t)^2 + (m_{b_2, \bar{\nu}, \ell^-} - m_t)^2 \quad (5.1)$$

where $m_t = 172.5$ GeV is the top quark mass. The assignment with the smaller χ^2 value is chosen.

ΔR -Method

This method assigns the b -jet candidates based on their angular distance to the leptons.

$$\Delta R = \sqrt{(\Delta\phi)^2 + (\Delta\eta)^2} \quad (5.2)$$

The jet-lepton-pair with the smaller ΔR value is assigned to each other. The remaining jet is assigned to the other lepton.

Note, that the χ^2 -Method relies on the reconstructed neutrino momenta to compute the invariant masses of the top quarks. Therefore, it is inherently linked to the neutrino regression method used. In contrast, the ΔR -Method is independent of the neutrino momenta. Further, it should be noted, that due to its reliance on the neutrino reconstruction and the use of ν^2 -Flows here, the χ^2 -method should also be considered a machine learning based approach.

The ΔR -Method was used in previous ATLAS analyses [1] for events, that failed the kinematic reconstruction required for the χ^2 -Method. However, in this thesis, both methods are evaluated on the full dataset for comparison.

5.2.3. Neutrino Regression Methods

Due to the limited availability of kinematic reconstruction methods in the ATLAS software framework, only one method for neutrino regression has been implemented and evaluated in this thesis.

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events

The baseline method used here is called ν^2 -Flows [45] and uses normalizing flows to model the conditional probability distribution of the neutrino momenta given the observed event kinematics. The model is trained on simulated dileptonic $t\bar{t}$ events, where the true neutrino momenta are known from the parton-level information. The trained model can then be used to sample possible neutrino momenta for new events, allowing for a probabilistic reconstruction of the event kinematics. For each event, multiple samples of neutrino momenta are drawn from the model, and the one with the highest probability density is selected as the reconstructed neutrino momenta. A detailed outline of the ν^2 -Flows method and its implementation can be found in Reference [45]. For this thesis, this method serves to establish a baseline for neutrino regression to evaluate the impact of improved jet assignment methods. Note, however, that this method has not been used in ATLAS analyses yet. However, due to matters of software compatibility, it was the only viable option at this stage of the thesis. Since ν^2 -Flows processes the full event information to make a prediction for the neutrino momenta, it does not require a jet-assignment. This is drastically different from the conventional methods, which were used in previous ATLAS analyses [1]. These methods are interdependent with the jet assignment and mostly based on exploiting the kinematic constraints of the event, such as the invariant masses of the W bosons and top quarks outlined in Section 2.2.3.

6. ML-Based Reconstruction of Dileptonic Top-Quark Pair Events

In addition to the outline reconstruction methods described in Chapter 5, this chapter focuses on the development of a machine learning-based reconstruction strategies for the dileptonic $t\bar{t}$ channel. Machine learning-based reconstruction techniques have gained significant traction in recent years. One major example for this, is their use in flavour tagging algorithms [68]. For this, machine learning models are trained to address both challenges of jet assignment and neutrino regression, by learning from simulated data to predict the correct jet-parton assignments and neutrino momenta.

For the high permutation multiplicity of jet assignments in all-hadronic channels, SPANet [6] has demonstrated significant improvements over traditional methods. These architectures leverage attention mechanisms to improve the accuracy by leveraging the full event information and utilising permutation equivariant architectures to handle the combinatorial nature of the problem. Based on these successes, this thesis explores the adaptation of similar architectures for the dileptonic $t\bar{t}$ channel, which presents unique challenges due to the presence of two neutrinos and fewer jets. This topic has been largely understudied in the literature, with most efforts focusing on neutrino regression. Thus, this work aims to fill this gap by developing a machine learning model specifically tailored for the jet assignment problem in the dileptonic channel, and investigating how it can be combined with neutrino regression methods to offer a full reconstruction pipeline.

On the other hand, for neutrino regression, ν^2 -Flows [45] has shown promise in improving the reconstruction of neutrino momenta by modelling the underlying probability distributions of the neutrino kinematics.

Despite both tasks being highly interconnected, both of them are mostly studied independently and the potential synergies between them remain largely unexplored. This thesis investigates how the jet assignment and neutrino regression tasks can be combined in a unified machine learning framework, allowing for a more holistic approach to event reconstruction in the dileptonic $t\bar{t}$ channel.

6.1. Jet-Assignment

6.1.1. Neural Network Architectures

6.1.2. Training Procedure

6.1.3. Hyperparameter Optimisation

6.2. Combination with Neutrino Regression

6.2.1. Unified Model Architecture

6.2.2. Training Strategy

6.2.3. Performance Evaluation

6.2.4. Multitask Learning Benefits

7. Performance-Evaluation of the ML-Based Reconstruction

7.1. Assignment Accuracy

7.2. Variable Reconstruction

7.3. Region Specific Performance

8. Data-Driven Validation Studies

8.1. Run 2 Data Sample

8.2. Analysis Setup

8.3. Results

9. Conclusion

9.1. Summary

9.2. Outlook

A. Additional Plots

B. Supplementary Computations and Proofs

B.1. Mathematical Foundation of Machine Learning

B.1.1. Mean Squared Error Loss and Conditional Expectation

When performing a minimisation of the mean squared error (MSE) loss function in a supervised learning setting, one can show that the optimal predictor corresponds to the conditional expectation of the target variable given the input features. The MSE loss function can be written as:

$$L_{\text{MSE}}(\mathbf{y}, f_{\theta}(\mathbf{x})) = \frac{1}{N} \sum_{i=1}^N (y_i - f_{\theta}(\mathbf{x}_i))^2 \quad (\text{B.1})$$

Using a simple calculus argument, one can show that the function f_{θ} that minimises this loss function is given by:

$$\frac{d}{d\theta} L_{\text{MSE}}(\mathbf{y}, f_{\theta}(\mathbf{x})|\mathbf{x}) = \frac{d}{df} \int (y - f_{\theta}(x))^2 p(y|x) dy \cdot \frac{\partial f}{\partial \theta} \quad (\text{B.2})$$

$$= \int 2(f_{\theta}(x) - y) p(y|x) dy \cdot \frac{\partial f}{\partial \theta} \stackrel{!}{=} 0 \quad (\text{B.3})$$

$$\Rightarrow f_{\theta}(x) = \int y p(y|x) dy = \mathbb{E}[y|x] \quad (\text{B.4})$$

Bibliography

- [1] ATLAS Collaboration, *Observation of quantum entanglement with top quarks at the ATLAS detector*, Nature **633**, 542 (2024)
- [2] ATLAS Collaboration, *Observation of a cross-section enhancement near the $t\bar{t}$ production threshold in $\sqrt{s} = 13$ TeV pp collisions with the ATLAS detector* (2026), 2601.11780, URL <https://arxiv.org/abs/2601.11780>
- [3] CMS Collaboration, *Observation of a pseudoscalar excess at the top quark pair production threshold*, Rep. Prog. Phys. **88**, 087801 (2025)
- [4] M. Jezabek et al., *V - A tests through leptons from polarised top quarks*, Phys. Lett. B **329**(2), 317 (1994)
- [5] A. Brandenburg et al., *QCD-corrected spin analysing power of jets in decays of polarized top quarks*, Phys. Lett. B **539**(3), 235 (2002)
- [6] A. Shmakov et al., *SPANet: Generalized permutationless set assignment for particle physics using symmetry preserving attention*, SciPost Phys. **12**, 178 (2022)
- [7] Particle Data Group Collaboration, *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024)
- [8] S. L. Glashow, *Partial Symmetries of Weak Interactions*, Nucl. Phys. **22**, 579 (1961)
- [9] S. Weinberg, *A Model of Leptons*, Phys. Rev. Lett. **19**, 1264 (1967)
- [10] A. Salam, *Weak and Electromagnetic Interactions*, Conf. Proc. C **680519**, 367 (1968)
- [11] D. J. Gross et al., *Asymptotically Free Gauge Theories*, Phys. Rev. D **8**, 3633 (1973)
- [12] H. D. Politzer, *Reliable Perturbative Results for Strong Interactions*, Phys. Rev. Lett. **30**, 1346 (1973)
- [13] G. 't Hooft, *Renormalizable Lagrangians For Massive Yang-Mills Fields*, Nucl. Phys. B **35**, 167 (1971)
- [14] P. W. Higgs, *Broken Symmetries, Massless Particles and Gauge Fields*, Phys. Lett. **12**, 132 (1964)
- [15] F. Englert et al., *Broken Symmetry and the Mass of Gauge Vector Mesons*, Phys. Rev. Lett. **13**, 321 (1964)

Bibliography

- [16] G. S. Guralnik et al., *Global Conservation Laws and Massless Particles*, Phys. Rev. Lett. **13**, 585 (1964)
- [17] T. D. Lee et al., *Question of Parity Conservation in Weak Interactions*, Phys. Rev. **104**, 254 (1956)
- [18] C. S. Wu et al., *Experimental Test of Parity Conservation in Beta Decay*, Phys. Rev. **105**, 1413 (1957)
- [19] N. Cabbibo, *Unitary Symmetry and Leptonic Decays*, Phys. Rev. Lett. **10**, 531 (1963)
- [20] M. Kobayashi et al., *CP Violation in the Renormalizable Theory of Weak Interaction*, Prog. Theor. Phys. **49**, 652 (1973)
- [21] H. D. Politzer, *Asymptotic Freedom: An Approach to Strong Interactions*, Phys. Rept. **14**, 129 (1974)
- [22] J. C. Collins et al., *Heavy Particle Production in High-Energy Hadron Collisions*, Nucl. Phys. B **263**, 37 (1986)
- [23] T. Sjöstrand, *A model for initial state parton showers*, Phys. Lett. B **157(4)**, 321 (1985)
- [24] G. Marchesini et al., *Simulation of QCD jets including soft gluon interference*, Nucl. Phys. B **238(1)**, 1 (1984)
- [25] T. Sjöstrand et al., *PYTHIA*, Comput. Phys. Commun. **135**, 238 (2001)
- [26] B. Andersson et al., *Parton fragmentation and string dynamics*, Phys. Rep. **97(2)**, 31 (1983)
- [27] T. Sjöstrand, *Jet fragmentation of multiparton configurations in a string framework*, Nucl. Phys. B **248(2)**, 469 (1984)
- [28] J. J. Aubert et al., *Experimental Observation of a Heavy Particle J*, Phys. Rev. Lett. **33**, 1404 (1974)
- [29] F. Abe et al. (CDF), *Observation of Top Quark Production in $p\bar{p}$ Collisions with the Collider Detector at Fermilab*, Phys. Rev. Lett. **74**, 2626 (1995)
- [30] S. Abachi et al. (DØ), *Observation of the Top Quark*, Phys. Rev. Lett. **74**, 2632 (1995)
- [31] I. Bigi et al., *Production and decay properties of ultra-heavy quarks*, Phys. Lett. B **181(1)**, 157 (1986)
- [32] G. Mahlon et al., *Spin correlation effects in top quark pair production at the LHC*, Phys. Rev. D **81**, 074024 (2010)
- [33] E. Schrödinger, *Discussion of Probability Relations between Separated Systems*, Math. Proc. Camb. Philos. Soc. **31(4)**, 555 (1935)

Bibliography

- [34] A. Peres, *Separability Criterion for Density Matrices*, Phys. Rev. Lett. **77**, 1413 (1996)
- [35] W. K. Wootters, *Entanglement of Formation of an Arbitrary State of Two Qubits*, Phys. Rev. Lett. **80**, 2245 (1998)
- [36] Y. Afik et al., *Entanglement and quantum tomography with top quarks at the LHC*, Eur. Phys. J. Plus **136(9)**, 907 (2021)
- [37] Y. Kiyo et al., *Top-quark pair production near threshold at LHC*, Eur. Phys. J. C **60**, 375 (2009)
- [38] Y. Sumino et al., *Bound-state effects on kinematical distributions of top quarks at hadron colliders*, J. High Energy Phys. **2010(9)**, 34 (2010)
- [39] K. Hagiwara et al., *Bound-state effects on top quark production at hadron colliders*, Phys. Lett. B **666(1)**, 71 (2008)
- [40] B. Fuks et al., *Signatures of toponium formation in LHC run 2 data*, Phys. Rev. D **104**, 034023 (2021)
- [41] B. Fuks et al., *Simulating toponium formation signals at the LHC*, Eur. Phys. J. C **85(2)**, 157 (2025)
- [42] L. Sonnenschein, *Analytical solution of $t\bar{t}$ dilepton equations*, Phys. Rev. D **73**, 054015 (2006)
- [43] V. Abazov et al., *Measurement of the top quark mass in the dilepton channel*, Phys. Lett. B **655(1)**, 7 (2007)
- [44] V. M. Abazov et al. (D0 Collaboration), *Measurement of the top quark mass in final states with two leptons*, Phys. Rev. D **80**, 092006 (2009)
- [45] J. A. Raine et al., *Fast and improved neutrino reconstruction in multineutrino final states with conditional normalizing flows*, Phys. Rev. D **109**, 012005 (2024)
- [46] W. Bernreuther et al., *Top quark pair production and decay at hadron colliders*, Nucl. Phys. B **690**, 81 (2004)
- [47] W. Bernreuther et al., *A set of top quark spin correlation and polarization observables for the LHC: Standard Model predictions and new physics contributions*, J. High Energy Phys. **2015(12)**, 1 (2015)
- [48] T. Gleisberg et al., *SHERPA 1.α, a proof-of-concept version*, J. High Energy Phys. **0402**, 056 (2004)
- [49] C. E. Shannon, *A mathematical theory of communication*, Bell Labs Tech. J. **27(3)**, 379 (1948)
- [50] M. Stone, *Cross-Validatory Choice and Assessment of Statistical Predictions*, Journal of the Royal Statistical Society: Series B (Methodological) **36(2)**, 111 (1974)

Bibliography

- [51] S. Kullback et al., *On information and sufficiency*, Ann. Math. Stat. **22(1)**, 79 (1951)
- [52] F. Rosenblatt, *The perceptron: A probabilistic model for information storage and organization in the brain.*, Psychol. Rev. **65(6)**, 386 (1958)
- [53] V. Nair et al., *Rectified linear units improve restricted boltzmann machines*, in *Proceedings of the 27th International Conference on International Conference on Machine Learning*, ICML'10, pages 807–814, Omnipress, Madison, WI, USA (2010)
- [54] D. E. Rumelhart et al., *Learning representations by back-propagating errors*, Nature **323(6088)**, 533 (1986)
- [55] J. L. Elman, *Finding structure in time*, Cogn. Sci. **14(2)**, 179 (1990)
- [56] S. Hochreiter et al., *Long Short-Term Memory*, Neural Comput. **9(8)**, 1735 (1997)
- [57] A. Vaswani et al., *Attention is All you Need*, in I. Guyon et al., editors, *Advances in Neural Information Processing Systems*, volume 30, Curran Associates, Inc. (2017)
- [58] J. Kiefer et al., *Stochastic Estimation of the Maximum of a Regression Function*, Ann. Math. Stat. **23(3)**, 462 (1952)
- [59] D. P. Kingma et al., *Adam: A Method for Stochastic Optimization*, International Conference on Learning Representations (ICLR) (2015)
- [60] N. Srivastava et al., *Dropout: A Simple Way to Prevent Neural Networks from Overfitting*, J. Mach. Learn. Res. **15(56)**, 1929 (2014)
- [61] I. Loshchilov et al., *Decoupled Weight Decay Regularization*, International Conference on Learning Representations (2019)
- [62] ATLAS Collaboration, *Luminosity determination in pp collisions at $\sqrt{s} = 13$ TeV using the ATLAS detector at the LHC*, ATLAS-CONF-2019-021 (2019)
- [63] ATLAS Collaboration, *High-Luminosity Large Hadron Collider (HL-LHC): Technical design report*, CERN Yellow Reports: Monographs, CERN, Geneva (2020)
- [64] ATLAS Collaboration, *The ATLAS Experiment at the CERN Large Hadron Collider*, J. Instrum. **3**, S08003 (2008)
- [65] P. Nason, *A New method for combining NLO QCD with shower Monte Carlo algorithms*, J. High Energy Phys. **11**, 040 (2004)
- [66] M. Cacciari et al., *The anti- k_t jet clustering algorithm*, J. High Energy Phys. **04**, 063 (2008)
- [67] ATLAS Collaboration, *Transforming jet flavour tagging at ATLAS*, Nat. Commun. **17(1)**, 541 (2026)
- [68] S. Mondal et al., *Machine learning in high energy physics: a review of heavy-flavor jet tagging at the LHC*, Eur. Phys. J. Spec. Top. **233(15)**, 2657 (2024)